



- 1 **RF TX and RX amplifiers:** These handle the signals in an out of the antenna.
- 2 **The inner circuit:** A lot of chips are placed on the PCB. Each has an assigned function.
- 3 **RF and Power:** It handles the power management and recharging of the cell phone; it also deals with the reception of FM channels in phones with that feature.
- 4 **Microprocessor and memory:** Both these chips co-ordinate the various functions of the cell phone and are responsible for communicating with the base station
- 5 **DSP:** The Digital Signal Processor is a customised processor that performs signal manipulation calculations at high speeds. It also takes care of signal compression and decompression.
- 6 **D/A and A/C chips:** These chips work in conjunction with each other and are responsible for the conversion of analogue signals to digital and vice versa.

Facts

- A message comprising 0 to 160 characters is considered a single SMS. So your service provider will charge you for three SMSes if your message comprises 450 characters.
- Tri-Band phones give you the convenience of communicating across five continents. However, make sure your service provider has tie-ups with international networks to implement this facility.
- Depending upon your service provider, a roaming scheme may not include outgoing calls from all states.
- Use only approved accessories and batteries. Do not connect incompatible products. For example, do not charge your phone with a charger that has a power rating higher than the one specified for that model. It might lower the battery life, or even damage the phone.



Buying Tips

- ✓ **BATTERY LIFE:** Most manufacturers claim to have a 15-day standby time under test conditions. Practically, that is never the case. Hence, do not base your buying decision solely on this.
- ✓ **TALK-TIME:** Always check the talk-time before making a purchase. Most mobile manufacturers specify this on the box itself.
- ✓ **SUPPORT & SERVICE:** What happens when your phone breaks down? Does the warranty cover the critical parts such as circuit board, screen, software installation and BIOS? Get this verified before paying up.
- ✓ **GENERAL PACKET RADIO SERVICE (GPRS):** Opt for a phone that supports WAP over GPRS if you intend to surf the Internet and access e-mail.

Quick Find	
YOU NEED	To talk on the move
Basic Use	
LOOK FOR	A phone that allows for 100 or more phone book entries
YOU NEED	An all-round communicator
Inter Use	
LOOK FOR	A phone with a storage capacity of 500 names and 150 SMSes, GPRS, infra-red or Bluetooth connectivity
YOU NEED	A high powered gadget
Prof Use	
LOOK FOR	A phone with a camera that supports a resolution of 640 x 480, loads of internal memory, and a fast CPU

The Future

Mobile phones are evolving fast. It's no wonder then that today's latest handset model will be relegated to the sidelines within the next few months. Network technologies are moving just as fast. We will soon be able to use 'third generation' (3G) frequency networks that offer faster data access. Chances are that this trend may lead to a huge switchover from landline connections to cellular phones. However, the huge price paid for the licenses suggests that these facilities won't come cheap.

Another interesting development in the mobile phone market is the disposable cell phone. The phone will be extremely small (about the thickness of three credit cards) and lightweight. Better still is the price—they are expected to sell for under \$20—a price that includes 60 minutes of pre-paid calling time.